

BLACKVILLE 3 W, SC, USA

WMO: 747840

Lat: 33.3550N

Lon: 81.3279W

Elev: 97

StdP: 100.17

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.4	-2.4	-13.2	1.2	-1.3	-10.6	1.5	1.1	6.7	8.5	6.2	9.8	1.2	N/A	0.326

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.8	24.0	33.6	23.9	32.5	23.7	25.8	31.3	25.4	30.9	25.0	30.4	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	19.6	28.2	23.9	19.0	27.8	23.4	18.5	27.3	80.2	31.4	78.0	30.9	76.2	30.6	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
5.6	4.7	4.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-7.9	36.4	2.0	1.8	-9.4	37.7	-10.6	38.8	-11.7	39.8	-13.2	41.1
			-8.7	26.9	2.4	0.5	-10.5	27.3	-11.9	27.6	-13.3	27.9	-15.1	28.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	7.8	9.7	13.4	17.6	22.1	25.7	26.8	26.3	23.8	18.5	12.5	9.6
	DBStd	7.82	5.60	5.01	4.76	3.89	3.15	2.07	1.75	1.91	2.56	4.05	4.47	4.99
	HDD10.0	292	111	62	23	2	0	0	0	0	0	1	23	70
	HDD18.3	1302	326	243	163	59	8	0	0	0	1	48	181	272
	CDD10.0	3168	44	54	127	230	376	472	522	506	416	264	99	59
	CDD18.3	1138	1	3	9	37	126	222	264	247	167	53	6	3
	CDH23.3	9947	4	16	92	379	1134	2047	2487	2107	1246	389	39	7
	CDH26.7	4032	0	0	9	84	416	918	1147	906	462	88	2	0
Wind	WSAvg	1.7	2.0	2.1	2.1	1.9	1.7	1.5	1.4	1.2	1.4	1.5	1.6	1.8
Precipitation	PrecAvg	1209	115	100	115	80	91	138	120	123	90	73	62	87
	PrecMax	1908	212	212	247	330	263	293	243	308	256	373	204	205
	PrecMin	844	18	18	25	9	18	17	31	24	3	0	18	21
	PrecStd	207	49	47	59	57	54	66	54	65	60	70	41	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	25.6	28.0	30.6	34.5	35.8	36.0	35.9	34.1	31.0	26.8	24.5
		MCWB	17.4	18.1	17.7	19.9	21.7	24.1	24.5	24.2	22.7	23.3	19.2	20.1
	2%	DB	21.1	22.8	25.8	29.0	32.3	34.3	34.7	34.0	32.3	29.0	24.3	22.1
		MCWB	16.0	16.6	17.4	19.0	21.3	23.8	24.5	24.5	23.0	21.3	17.5	18.1
	5%	DB	19.1	20.5	24.0	27.4	30.8	32.9	33.5	32.7	31.0	27.3	22.3	20.1
		MCWB	15.0	15.2	16.2	18.7	21.0	23.4	24.4	24.2	22.5	20.3	16.4	17.1
	10%	DB	17.0	18.3	21.9	25.6	29.1	31.6	32.2	31.4	29.6	25.5	20.5	18.2
		MCWB	13.7	14.3	15.4	17.8	20.6	23.4	24.4	24.0	22.5	19.3	16.2	15.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	19.7	20.0	22.5	23.8	25.8	26.7	26.6	25.4	24.7	20.8	21.1
		MCDB	21.4	23.3	24.7	27.6	29.4	31.9	32.8	31.6	30.1	28.3	23.3	23.1
	2%	WB	17.7	18.4	18.9	21.3	22.9	25.2	25.8	25.8	24.7	23.5	19.5	19.7
		MCDB	20.7	22.3	23.8	25.7	28.4	31.7	31.8	30.7	29.4	27.0	21.7	21.5
	5%	WB	16.1	17.2	17.9	20.5	22.4	24.7	25.4	25.3	24.0	22.5	18.5	17.9
		MCDB	18.2	20.6	22.3	24.8	27.7	31.0	31.4	30.2	28.7	25.6	21.0	19.5
	10%	WB	14.3	15.6	16.7	19.5	21.8	24.1	25.0	24.9	23.4	21.2	16.9	16.2
		MCDB	16.2	18.6	20.7	23.9	27.0	30.2	30.7	29.6	27.7	24.1	19.7	18.2
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.7	12.4	12.8	11.9	10.9	10.6	9.9	10.3	11.7	12.3	11.0
		MCDBR	12.7	13.6	14.4	14.7	14.2	12.8	12.1	11.8	12.4	12.7	13.7	11.6
		MCWBR	8.4	8.0	7.2	6.2	5.2	4.4	4.1	4.1	4.6	6.1	8.0	7.6
	5% WB	MCDBR	10.3	11.5	11.8	10.9	11.0	11.2	10.7	9.9	10.4	10.0	10.4	9.3
		MCWBR	8.0	8.0	6.7	5.8	4.5	4.4	4.1	4.1	4.4	5.7	7.7	7.5
Clear-Sky Solar Irradiance	taub	0.324	0.332	0.367	0.403	0.470	0.508	0.550	0.526	0.437	0.400	0.347	0.320	
	taud	2.487	2.463	2.393	2.285	2.142	2.082	1.986	2.042	2.280	2.345	2.460	2.535	
	Ebn at Noon	892	920	909	886	827	792	756	768	828	831	855	876	
	Edh at Noon	87	98	114	132	155	164	179	167	126	109	89	79	
All-Sky Solar Radiation	RadAvg	2.84	3.47	4.54	5.73	6.18	6.34	6.09	5.54	4.79	4.05	3.11	2.48	
	RadStd	0.15	0.35	0.34	0.40	0.45	0.44	0.42	0.29	0.41	0.47	0.23	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+228	+151
(47)	Regional (53 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+131

Nomenclature: See separate page